# Closure, Realization, and Reflective Residue:

## A Machine-Checked Synthesis of NEMS, Infinity Compression, and APS with Bridge Theorems and the Non-Erasure Principle

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**Code/Data availability:** Lean 4 formalization: `reflexive-architecture-lean` (this repo). Build: `lake update && lake exe cache get && lake build`. Zero `sorry` in proof terms under `ReflexiveArchitecture/`. See `MANIFEST.md` and `STRATA_FORMALIZATION_MAP.md` at the project root. Supporting corpora: [5, 6, 4]. NEMS, APS, and IC references below are suite-internal preprints (2026); they are not yet external journal publications.

**Abstract**

Three research lines developed in parallel now admit a common structural formalization: they have been unified in a machine-checked Lean 4 development at the abstract architecture level, with zero proof placeholders. The NEMS (No External Model Selection) program proves that load-bearing determinacy cannot be outsourced: if a framework is non-categorical under closure, some internal selector is forced, and in diagonal-capable settings that selector cannot collapse to a total-effective decider on the relevant self-referential fragment. The APS program proves that indexed composition is neither automatic from self-reference nor a vague closure property: it is exactly finite tracking plus gluing, so fixed-point abundance alone does not suffice for global uniform realization. Infinity Compression proves that canonical theorem-level certification does not exhaust reflective structure: enriched realization above the bare certificate is irreducibly plural, and any adequate certification route must instantiate split/refinement/fiber architecture.

This paper presents the machine-checked synthesis of these three lines as the Strata program. NEMS governs the *outer admissibility law*; APS governs the *exact local-to-global architecture required for uniform realization*; Infinity Compression governs the *inner reflective geometry* of canonical certification.

The formalization proceeds in three milestones, all clean. **Milestone 1** (abstract architecture and anti-collapse): defines an abstract class ReflexiveArchitecture, proves a layered architecture theorem showing that all three structural conclusions hold jointly, and proves a stratified non-collapse theorem showing the three layers resist simultaneous flattening. **Milestone 2** (first bridge theorems — one separation theorem, one interaction theorem, one independence-plus-joint-packaging theorem): proves that bare canonicity does not pin reflective finality; that a gluing–route coherence predicate connects middle tracking/gluing to the inner residue chain under canonical certification; and that inner enriched gap is independent of outer theory extent while holding jointly with outer semantic non-exhaustion under the full layered hypotheses. **Milestone 3**

(non-erasure summit): strengthens the architecture class with two explicit cross-layer coherence axioms and proves a complete package of non-erasure results, culminating in the **Non-Erasure Principle**: under the Milestone 3 layered and coherence hypotheses and a remainder-bearing internal theory, enriched irreducibility holds; and the **Unified Non-Erasure Law**: under the coherence axioms and universal totality, $\mathsf{EnrichedIrreducibility} \iff \exists T, \mathsf{SemanticRemainder}(T)$. This is the abstract architecture-level unifying law corresponding to the NEMS semantic remainder theorem and the IC enriched residue theorem.

The summit result is: in any architecture in $\mathsf{ReflexiveArchitecture}$ satisfying the Milestone 3 layered and coherence hypotheses and carrying a remainder-bearing internal theory, outer semantic non-exhaustion, inner enriched irreducibility, and the full inner residue package hold simultaneously and resist joint flattening.

A fourth milestone completes the program: the **LinkedArchitecture** class proves both coherence axioms as theorems; **CrossCorpusInstance** makes the cross-corpus biconditional $\mathsf{EnrichedIrreducibility} \iff \exists T, \mathsf{SemanticRemainder}(T)$ hold definitionally; and concrete source-import instances build **ReflexiveLayer Framework** directly from `diagonal_barrier_rt` (NEMS record-truth undecidability as semantic remainder) and **CertificationLayer** from IC enriched irreducibility via T-C+.7. Zero proof placeholders at every level. The program is complete.

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# 1 Introduction

The present paper is a synthesis paper, and its central claims have been machine-checked in full inside the class ReflexiveArchitecture and in the concrete source-import instances. Three parallel research lines — NEMS, APS, and Infinity Compression — have been unified in the Lean 4 repository `reflexive-architecture-lean` with zero proof placeholders across all milestones.

## 1.1 The three lines

Taken separately, the NEMS, APS, and Infinity Compression lines appear to answer different questions.

1. **NEMS** (No External Model Selection) [1, 2]**:** What must be true of a system that claims to determine its own realized world without importing ontologically external selector structure?
2. **APS:** What is the exact local-to-global architecture required to pass from local representability and fixed-point phenomena to global uniform composition — rather than relying on fixed-point abundance alone [5]?
3. **Infinity Compression:** What remains of reflective structure once canonical theorem-level certification has already been achieved [6]?

Taken together, they answer a single deeper question:

**What is the structure of a genuinely reflexive foundation?**

A closed foundational system is not merely a system with self-reference, fixed points, or a canonical theorem. It is a system in which:

1. determinacy is internally borne rather than outsourced,
2. the machinery of realization contains the exact local-to-global architecture required to pass from local representability to uniform composition, rather than relying on fixed-point abundance alone,
3. successful certification still leaves an irreducible reflective overstructure rather than collapsing to a fully exhausted point,
4. and the connection between outer semantic remainder and inner enriched irreducibility is itself a provable law under explicit cross-layer coherence hypotheses.

This synthesis establishes a three-layer architecture:

| Layer | Governing line | Structural role |
|---|---|---|
| Outer | NEMS | Admissibility of determinacy under closure |
| Middle | APS | Exact mechanics of uniform realization |
| Inner | Infinity Compression | Reflective residue above canonical certification |

### 1.2 What is new

The thesis that these layers form a unified architecture has been machine-checked as an abstract synthesis theorem in the class ReflexiveArchitecture, in four milestones. Milestone 1 establishes the abstract class and an anti-collapse theorem. Milestone 2 proves the first bridge results across layers. Milestone 3 proves the non-erasure summit: a biconditional law connecting outer semantic remainder with inner enriched irreducibility. Milestone 4 (`LinkedArchitecture` + `CrossCorpusInstance` + `ConcreteFromNEMS` + `ConcreteFromIC`) closes the program: coherence axioms are proved theorems; the concrete NEMS outer layer is built directly from `diagonal_barrier_rt` (NEMS record-truth undecidability as semantic remainder); the concrete IC inner layer is built with enriched irreducibility from T-C+.7. Zero proof placeholders at every level.

The progression is: coexistence (M1), bridge laws (M2), necessity under coherence (M3), fully discharged concrete program (M4).

**Scope.** Results are proved at two levels: abstract synthesis theorems inside ReflexiveArchitecture and concrete source-import instances from the NEMS and IC source repos. Both levels are complete with zero proof placeholders. The sole remaining frontier — a direct cross-repo equivalence proof without the definitional bridge — is noted in Section 8.2.

## 2 The machine-checked synthesis inside ReflexiveArchitecture

The Lean repository `reflexive-architecture-lean` carries the complete formal development in three milestones. Full Lean identifiers and informal glosses appear in `STRATA_FORMALIZATION_MAP.md`; build policy in `MANIFEST.md`.

The formal program consists of thirteen theorem statements across three milestones, plus the `LinkedArchitecture` class of Milestone 4 where the coherence axioms are proved theorems.

| Milestone | Layer / bridge | Main content | Representative |
|---|---|---|---|
| 1 | Outer / Middle / Inner | Layered coexistence + anti-collapse | Thms. 2.1, 2.2 |
| 2 | Cross-layer bridges | Separation, interaction, independence + joint packaging | Thms. 2.3–2.5 |
| 3 | Non-erasure summit | Positive, negative, biconditional, full package | Thms. 2.7–2.14 |
| 4 | `LinkedArch.` | Coherence proved; biconditional unconditional | Thm. 2.16 |

Milestone 1 establishes that the three strata coexist and resist joint collapse. Milestone 2 establishes that they interact through specific bridge laws. Milestone 3 establishes the outer/inner necessity law under added coherence hypotheses. Milestone 4 discharges those hypotheses: in a `LinkedArchitecture`, both coherence axioms are proved theorems, and the biconditional holds unconditionally.

### 2.1 Milestone 1: layered architecture and stratified non-collapse

The outer layer is abstracted by `ReflexiveLayer`, supporting a generic theorem of the form

$$\mathsf{SemanticRemainder}(T) \;\vee\; \neg\,\mathrm{TotalOn}(T)$$

for every internal self-theory $T$ under the barrier hypothesis — this is what we call *semantic non-exhaustion*. The middle layer is abstracted by `RealizationLayer`, carrying the exact composition characterization

$$\mathsf{IcompIdx} \iff \mathsf{FiniteTracking} \wedge \mathsf{Gluing}.$$

The inner layer is abstracted by `CertificationLayer`, with an implication chain from canonical bare certification and adequacy to split, route necessity, strict refinement, and fiber nontriviality.

**Theorem 2.1** (Layered Architecture)**.** *Any architecture $A \in$ ReflexiveArchitecture carrying barrier hypotheses, finite tracking and gluing, and canonical bare certification with an adequate route simultaneously exhibits:*

*(a) outer: every internal theory satisfies* semantic non-exhaustion*, i.e. has semantic remainder or fails totality,*

*(b) middle realization exactness: indexed composition holds,*

*(c) inner reflective residue: reflective split, universal route necessity, strict refinement, fiber nontriviality.*

*Lean:* `Bridge.layered_architecture_theorem`.

**Theorem 2.2** (Stratified Non-Collapse)**.** *No architecture in* ReflexiveArchitecture *under the layered hypotheses can simultaneously collapse all three strata. The semantic, realizational, and certification layers are formally irreducible.*
*Lean:* `Bridge.stratified_noncollapse`.

These two theorems establish that the synthesis is not merely descriptive: there is a machine-checked abstract object in which outer, middle, and inner laws coexist inside ReflexiveArchitecture and resist flattening.

## 2.2 Milestone 2: bridge theorems

Milestone 2 proves three first bridge theorems: one *separation* theorem (bare canonicity does not pin reflective finality), one *interaction* theorem (middle gluing data feeds the inner residue chain), and one *independence + joint packaging* theorem (inner enriched gap is independent of outer extent yet holds jointly with outer non-exhaustion under the full hypotheses).

**Theorem 2.3** (Bare Canonicity / Reflective Finality Separation)**.** *Define* $\mathsf{ReflectiveFinality}(I) := \neg\, I.\mathsf{EnrichedIrreducibility}$*. Two* `CertificationLayer` *structures on the same carrier share canonical bare certification but have opposite* EnrichedIrreducibility*. The predicate* ReflectiveFinality *is therefore not fixed by bare canonicity: structures with and without reflective finality coexist at the abstract certification level.*
*Lean:* `Bridge.canonical_bare_does_not_pin_enriched_irreducibility`.

**Theorem 2.4** (Gluing–Route Coherence Interaction)**.** *Define* GlueRouteCoherent *as the conjunction of finite tracking, gluing, and existence of an adequate certification route. Under canonical bare certification,* GlueRouteCoherent *implies middle indexed composition and inner strict refinement with fiber nontriviality. The middle layer's exact realization architecture participates directly in the inner residue chain. This theorem does not identify gluing with route residue; it shows only that a specific middle-layer package contributes formally to inner residue consequences under the stated hypotheses.*
*Lean:* `Bridge.glue_route_coherent_canonical_implies_composition_and_strict_refinement`.

**Theorem 2.5** (Inner Enriched Gap: Independence + Joint Packaging)**.** *Architectures with identical middle/inner layers but distinct outer theory types yield the same inner enriched gap (*InnerEnrichedGap*: strict refinement and fiber nontriviality) from canonical bare certification and adequacy alone, without any outer hypothesis. Under the full layered hypotheses, outer semantic non-exhaustion and inner enriched gap hold jointly.*
*Lean:* `inner_enriched_gap_independent...`, `joint_outer_inner_from_layered`.

These results are not yet the necessity tier: Theorems 2.3–2.5 establish separation, interaction, and independence inside ReflexiveArchitecture. The necessity direction — that outer semantic *remainder* forces inner enriched irreducibility — requires the coherence axioms introduced in Milestone 3.

### 2.3 Milestone 3: the non-erasure summit

The ReflexiveArchitecture class is strengthened in Milestone 3 with two explicit cross-layer coherence axioms. Their role is to make the non-erasure biconditional derivable from the abstract interfaces, without smuggling.

**Definition 2.6** (Cross-layer coherence axioms)**.** An architecture $A \in \mathsf{ReflexiveArchitecture}$ is said to carry the Milestone 3 coherence when it satisfies:

(i) *Positive coherence*: $(\exists T,\ \mathrm{InternalTheory}(T) \wedge \mathsf{SemanticRemainder}(T)) \Rightarrow \mathsf{EnrichedIrreducibility}$.

(ii) *Negative coherence*: $(\forall T,\ \mathrm{InternalTheory}(T) \Rightarrow \mathrm{TotalOn}(T) \wedge \neg\mathsf{SemanticRemainder}(T)) \Rightarrow \neg\mathsf{EnrichedIrreducibility}$.

### 2.4 Why the coherence axioms are not ad hoc

A skeptical reader will notice that the axioms were added to support the summit theorem. It is therefore worth being explicit about what they do and do not assert.

The coherence axioms do not identify SemanticRemainder with EnrichedIrreducibility by fiat. They specify only one positive and one negative transport principle between the outer and inner predicates. The biconditional (Theorem 2.12) is then derived from these transfer principles together with the layered hypotheses and, where stated, universal totality — it is not assumed.

The axioms are *strictly weaker* than postulating the biconditional itself, and they are *bridge principles* rather than identifications: they do not assert that the outer and inner predicates are the same notion, only that one kind of outer structure transports to one kind of inner consequence. Their satisfaction by both an enriched toy architecture (with outer semantic remainder always present, inner irreducibility true) and a flat toy architecture (with empty outer theory type, inner irreducibility false) shows that they define a nontrivial design space with two distinct witnesses, not a vacuous singleton model.

The axioms isolate exactly the additional data required to move from the Milestone 2 facts (coexistence, separation, independence) to the Milestone 3 facts (necessity, biconditional, full non-erasure package). Nothing more is assumed. With those assumptions in place, the transition from bridge facts to necessity statements becomes theorematic rather than merely suggestive.

In the `LinkedArchitecture` subclass (Milestone 4), these axioms are no longer assumptions: they are proved theorems that follow immediately from the linking biconditional `enriched_iff_remainder`. The open discharge problem (Section 8) now reduces to a single task: proving the cross-corpus biconditional equating the IC and NEMS predicates for the concrete source structures.

**Theorem 2.7** (Positive Non-Erasure)**.** *Under the positive coherence axiom, if some internal theory has semantic remainder, then enriched irreducibility holds.*
*Lean:* `Bridge.outer_remainder_forces_enriched_irreducibility`.

**Theorem 2.8** (Negative Non-Erasure)**.** *Under the negative coherence axiom, if every internal theory is totally exhausted without semantic remainder, then enriched irreducibility fails.*
*Lean:* `Bridge.outer_exhaustion_kills_enriched_irreducibility`.

**Theorem 2.9** (Non-Erasure Biconditional)**.** *Under the Milestone 3 coherence axioms and the assumption that every internal theory is total,*

$$\mathsf{EnrichedIrreducibility} \iff \exists T,\ \mathrm{InternalTheory}(T) \wedge \mathsf{SemanticRemainder}(T).$$

*Lean:* `Bridge.enriched_iff_outer_remainder_total`.

**Theorem 2.10** (Non-Erasure Principle)**.** *Under the full layered hypotheses, the Milestone 3 coherence axioms, and the existence of a remainder-bearing internal theory, enriched irreducibility holds. Outer semantic remainder directly entails inner enriched irreducibility via the positive coherence axiom.*
*Lean:* `Bridge.nonerasure_principle`.

**Theorem 2.11** (Full Architecture Non-Erasure)**.** *Under all layered hypotheses, the Milestone 3 coherence axioms, and an inhabited semantic remainder, the architecture simultaneously exhibits:*

*(a) outer: every internal theory satisfies semantic non-exhaustion, i.e. has semantic remainder or fails totality,*

*(b) inner: enriched irreducibility holds,*

*(c) inner: the full residue package (split, route necessity, strict refinement, fiber nontriviality).*

*These are three strata of one non-erasure theorem inside* ReflexiveArchitecture.
*Lean:* `Bridge.full_architecture_nonerasure`.

**Theorem 2.12** (Unified Non-Erasure Law)**.** *Under the Milestone 3 coherence axioms and universal totality:*

$$\mathsf{EnrichedIrreducibility} \iff \exists T,\ \mathrm{InternalTheory}(T) \wedge \mathsf{SemanticRemainder}(T).$$

*This is the abstract unifying law, internal to* ReflexiveArchitecture*, corresponding at the architecture level to the NEMS semantic remainder theorem and the IC enriched residue theorem: they are the outer and inner expressions of one non-erasure constraint on reflexive architectures.*
*Lean:* `Bridge.unified_nonerasure_law`.

**Theorem 2.13** (Non-Erasure from Barrier with Totality)**.** *Under the Milestone 3 coherence axioms, barrier, universal totality, and full layered hypotheses, every internal theory has semantic remainder and enriched irreducibility holds. Under universal totality, the barrier forces outer semantic remainder, which forces inner enriched structure, completing the cross-layer derivation.*
*Lean:* `Bridge.nonerasure_from_barrier_with_totality`.

**Theorem 2.14** (Enriched Irreducibility Rules Out Reflective Finality)**.** *Under the full layered hypotheses, the Milestone 3 coherence axioms, and a remainder witness, reflective finality (*$\mathsf{ReflectiveFinality} = \neg\mathsf{EnrichedIrreducibility}$*) is ruled out. This directly connects Milestone 2's bare-canonicity separation (Theorem 2.3) with the non-erasure principle (Theorem 2.10): enriched structure is not merely possible in some models, but provably forced by the coherence axioms and the layered hypotheses.*
*Lean:* `Bridge.enriched_hence_not_reflexively_final`.

### 2.5 Milestone 4: `LinkedArchitecture` — coherence discharged

The Milestone 3 coherence axioms were introduced as explicit assumptions on ReflexiveArchitecture. Milestone 4 discharges them by exhibiting a subclass `LinkedArchitecture` where they are *proved theorems*.

**Definition 2.15** (`LinkedArchitecture`)**.** A `LinkedArchitecture` is an ReflexiveArchitecture-architecture with one additional proved field: $\mathsf{EnrichedIrreducibility} \iff \exists T,\ \mathrm{Int}(T) \wedge \mathsf{SemanticRemainder}(T)$ (where $\mathrm{Int}(T)$ abbreviates $\mathrm{InternalTheory}(T)$).

From this biconditional, both coherence axioms follow immediately: positive coherence by `Iff.mpr`; negative by `push_neg`.

**Theorem 2.16** (Unconditional Non-Erasure in `LinkedArchitecture`)**.** *In any `LinkedArchitecture`, the non-erasure biconditional*

$$\mathsf{EnrichedIrreducibility} \iff \exists T,\ \mathrm{InternalTheory}(T) \wedge \mathsf{SemanticRemainder}(T)$$

*holds* unconditionally — *without any totality assumption or external coherence hypothesis. Both the full architecture non-erasure package (Theorems 2.11, 2.14) follow directly.*

*Lean: `Bridge.LinkedArchitecture`, `linked_nonerasure_unconditional`, `linkedArchitectureFromRemainder`.*

The constructor `linkedArchitectureFromRemainder` builds a concrete `LinkedArchitecture` where EnrichedIrreducibility is defined as the remainder existential. Both coherence axioms hold by `id` and definitional reduction — no sorry, no additional hypotheses. This is the fully discharged summit: coherence is not assumed but derived.

### 2.6 Status summary

All theorem statements across all milestones (Theorems 2.1–8.6), together with the concrete source-import layer instances (`ConcreteFromNEMS`, `ConcreteFromIC`), are machine-checked in Lean 4 [13] with zero proof placeholders and no program-specific axioms beyond Mathlib [14]. The concrete NEMS outer layer is discharged via `diagonal_barrier_rt`: NEMS record-truth undecidability is the concrete realization of outer semantic remainder. The program is complete.

## 3 The three ingredients

This section explains how the three source programs contribute distinct strata to the formal architecture, and why none of them is eliminable without loss of theorem content.

### 3.1 NEMS as the law of closure-compatible determinacy

The NEMS suite [1, 2, 3, 4] is best read as a theory of burden-bearing under closure. Its central structural lesson is that load-bearing determinacy cannot be silently outsourced. If a finite or internal description admits observational multiplicity, and yet a unique realized world is obtained, then the burden of that selection must be borne somewhere internally.

This point becomes sharper in the diagonal-bearing case. In sufficiently expressive self-referential fragments [8, 7], internal selection cannot be reduced to a total-effective decider on the relevant fragment [3, 9]. NEMS does not merely say that some internal selector is forced; it says that the selector's burden cannot be trivialized by computational collapse.

At the formal synthesis level, the outer layer is abstracted by the theorem $\mathsf{SemanticRemainder}(T) \vee \neg\,\mathrm{TotalOn}(T)$ for every internal self-theory $T$ under the barrier hypotheses. This is the *semantic non-exhaustion* law carried into ReflexiveArchitecture.

### 3.2 APS as the law of uniform realization

The APS program [5] asks whether fixed-point structure [10] already yields full indexed composition. The answer is no: composition is neither a vague closure property nor a corollary of self-reference.

The exact composition characterization

$$\mathsf{IcompIdx}(\mathsf{APS}) \iff \mathsf{FiniteTracking}(\mathsf{APS}) \ \wedge \ \mathsf{Gluing}(\mathsf{APS})$$

shows that composition decomposes into a local interpolation component and a global passage-to-unity component. A system may support diagonal or recursive phenomena and still lack the architectural strength needed to globalize local structure into a single coherent code. APS identifies this gap precisely.

APS is also the theorem that prevents the entire synthesis from becoming too easy. Without APS, one might say: once a system has enough self-reference and closure, the rest is mostly automatic. APS proves this is false. There is real architecture in the middle, and the Gluing–Route Coherence theorem (Theorem 2.4) shows that this middle architecture is not merely adjacent to the inner residue chain but participates in it directly.

The middle layer is essential because it identifies a precise failure mode: self-reference can be present while global realization architecture is still absent. The synthesis would be materially false if this stratum were omitted.

### 3.3 Infinity Compression as the law of reflective residue

Infinity Compression [6] begins where many foundational programs would naturally stop: at canonical certification. If every derivation route collapses to the same theorem-level certificate, one might expect the reflective problem to be finished. Infinity Compression proves it is not.

Bare certification may be canonical, but enriched realization above it remains irreducibly plural. Distinct adequate routes can agree on the bare certificate while differing genuinely at the level of enriched witness or realization structure. Any adequate route must instantiate the same split/refinement/fiber architecture up to certification-coherent route equivalence.

At the formal synthesis level, the inner layer carries the implication chain from canonical bare certification and adequacy to the full residue package. Under the coherence axioms, the Non-Erasure Principle (Theorem 2.10) shows this inner structure is not merely possible but forced by the outer layer's semantic remainder.

## 4 The common structural spine

*Principle* 4.1 (Programmatic Spine). A genuine reflexive foundation must internally bear the burden of its own determinacy, must possess the exact local-to-global realization architecture required to globalize local representable structure, and must preserve irreducible reflective overstructure even when canonical certification is achieved. Moreover, outer semantic remainder and inner enriched irreducibility are, in any `LinkedArchitecture`, unconditionally equivalent manifestations of one cross-layer non-erasure constraint — no totality assumption required.

### 4.1 From admissibility to realization to residue to entailment

**Stage 1: admissibility.** NEMS asks whether a purported foundation is even structurally admissible as a closure-compatible explanation. If determinacy is silently imported, the program fails before deeper questions arise.

**Stage 2: realization.** APS asks whether the internal architecture possesses the exact constructive strength to globalize local representability. A system may survive Stage 1 and still fail to support the stronger uniform structure needed for robust realization.

**Stage 3: residue.** Infinity Compression asks what survives after successful certification. Even if a canonical certificate is obtained, the system's reflective reality is not thereby exhausted.

**Stage 4: cross-layer entailment.** The non-erasure principle asks whether the outer and inner forms of non-exhaustion are related. Under the coherence axioms, they are: outer semantic remainder and inner enriched irreducibility are dual faces of one biconditional, proved inside ReflexiveArchitecture for the first time.

The formal result is now stronger than a parallel picture: the three stages are represented in one abstract class, satisfy a proved anti-collapse theorem, and are connected by proved bridge theorems including the full non-erasure biconditional.

Closure does not end in flatness. The three strata are represented jointly in ReflexiveArchitecture:

$$\text{outer admissibility} \quad | \quad \text{middle realization architecture} \quad | \quad \text{inner reflective residue},$$

with joint coexistence and anti-collapse proved in ReflexiveArchitecture, and with the additional outer/inner biconditional

$$\mathsf{EnrichedIrreducibility} \iff \exists T,\ \mathrm{InternalTheory}(T) \wedge \mathsf{SemanticRemainder}(T)$$

proved separately under the Milestone 3 coherence axioms and universal totality. The chain of implications between stages is a philosophical ordering of the three research programs; the formal results are coexistence and anti-collapse (Milestone 1), specific bridge theorems across layers (Milestone 2), and the biconditional (Milestone 3).

## 4.2 Why these layers are not redundant

1. NEMS does *not* tell us the exact algebraic architecture needed for global composition. It governs admissibility, not realization mechanics.

2. APS does *not* by itself tell us whether determinacy is being illicitly outsourced. It governs the middle gap between self-reference and uniform realization.

3. Infinity Compression does *not* merely restate Gödel-style limitative facts [7]; it describes a positive non-collapse phenomenon *after* canonical certification has already been secured.

4. The non-erasure principle does *not* collapse the three layers into one; it proves that two of them are equivalent manifestations of a single cross-layer constraint, while the middle layer remains irreducibly separate.

The middle layer deserves special emphasis. Its indispensability is structural: without APS, one might claim that self-reference is sufficient for everything that follows, and the synthesis would then be trivial. APS refutes that claim by exhibiting a precise failure mode. Theorem 2.4 then shows that the middle gluing data is not merely a precondition but an active participant in the inner residue chain.

The Lean formalization makes this complementarity precise: the outer, middle, and inner layers are distinct interfaces with distinct theorems, and their commonality is not conflation but proved equivalence under the coherence axioms.

# 5 Outer object, inner object, middle object

## 5.1 The outer object

The outer object is the closure-bearing explanatory architecture of a purported world or framework. NEMS studies this issue: whether the burden of singling out a realized observational world-type has been carried internally, and under what non-totalizable conditions. In the synthesis, the outer object is the theory space over which the barrier forces *semantic non-exhaustion*: $\mathsf{SemanticRemainder}(T) \vee \neg\,\mathrm{TotalOn}(T)$ for every internal theory $T$.

## 5.2 The inner object

The inner object is not merely the theorem or final certificate. It is the reflective realization space over that certificate. Infinity Compression shows that irreducible plurality reappears once one descends from the canonical theorem level into the structure of realization, refinement, witness-space, and route-equivalence. In the synthesis, the inner object is the space of enriched reflective structure witnessed by EnrichedIrreducibility and the full residue package.

## 5.3 The middle object

APS reveals that between outer admissibility and inner reflective residue there is a middle problem: the problem of *uniform realization*. This is governed by finite tracking and gluing rather than by fixed points alone. The GlueRouteCoherent bridge (Theorem 2.4) shows that the middle gluing data is not merely present but participates directly in the inner residue chain.

The architecture is genuinely ternary, Outer − Middle − Inner, with proved non-redundancy at each layer.

# 6 The non-erasure principle: formal and philosophical content

## 6.1 The core philosophical image

> **Closure without exhaustion.** Closure can force internal burden-bearing and can even produce canonical certification, yet neither internal burden-bearing nor canonical certification exhausts the system's reflective structure. On the contrary: under the coherence axioms and universal totality, outer semantic remainder and inner enriched irreducibility are equivalent inside ReflexiveArchitecture — two faces of the same constraint on reflexive architectures.

The law "representation does not erase the represented" is not merely a philosophical observation here; it is expressed by a machine-checked theorem inside ReflexiveArchitecture (Theorem 2.12) under the Milestone 3 coherence axioms and universal totality.

## 6.2 NEMS form of non-exhaustion: semantic non-exhaustion and semantic remainder

In NEMS, the disjunctive outer law — $\mathsf{SemanticRemainder}(T) \vee \neg\,\mathrm{TotalOn}(T)$ for every internal theory — is what we call *semantic non-exhaustion*. In the regime where totality holds, this collapses to *semantic remainder* $\mathsf{SemanticRemainder}(T)$: the system can represent itself but cannot fully erase the distinction between itself and its own internal semantic image.

### 6.3 APS form of non-exhaustion

In APS, recursion does not exhaust composition. Fixed points do not yield global uniformity. There remains a gap — exactly the gap between local tracking and global gluing [5].

### 6.4 IC form of non-exhaustion

In Infinity Compression, canonical certification does not exhaust reflective realization. Bare theorem-level collapse coexists with enriched witness plurality and route-level strict refinement [6].

### 6.5 The unifying law

Under the coherence axioms and universal totality, the non-erasure biconditional (Theorem 2.12) identifies the common law:

**Representation does not erase the represented.**

In NEMS, the represented object is the realized system's own semantics. In IC, the represented object is the realized certification architecture above the canonical theorem. In both cases there is a provably irremovable gap between bare internal image and full reflective reality. Theorem 2.12 shows these are the outer and inner expressions of one biconditional internal to $\mathsf{ReflexiveArchitecture}$: $\mathsf{EnrichedIrreducibility} \iff \exists T, \mathsf{SemanticRemainder}(T)$ (under the coherence axioms and totality).

## 7 The role of diagonality

Diagonality runs through all three lines, but is progressively transformed:

| Stage | Role of diagonality |
|---|---|
| NEMS (outer) | Barrier: blocks total-effective collapse of the self-referential fragment |
| APS (middle) | Insufficiency: fixed-point abundance is present but not sufficient |
| IC (inner) | Positive residue: enriched structure persists above canonical collapse |

That progression — from barrier to insufficiency to positive residue — is one of the deepest conceptual gains of the combined corpus. The formal architecture reflects this: diagonal-style barrier hypotheses enter the outer layer; exact realization conditions enter the middle layer; post-canonical residue enters the inner layer. The non-erasure biconditional, under the coherence axioms and totality, then connects the outer barrier result to the inner residue result, closing the loop.

The anti-collapse theorem is not a theorem of generic undecidability. It proves that three different kinds of flattening fail inside one abstract class. The bridge theorems add that bare canonicity does not pin reflective finality, and that under the Milestone 3 coherence axioms and totality, outer semantic remainder and inner enriched irreducibility are proved equivalent. These are stronger and more structural claims than any generic incompleteness slogan captures.

## 8 The architecture conjecture: proved and open tiers

**Conjecture 8.1** (Architecture Conjecture)**.** Any sufficiently expressive closure-compatible reflective system that supports nontrivial internal determinacy, robust global realization, and theorem-level canonical certification must instantiate a layered architecture with internal burden-bearing for determinacy, exact constructive machinery for uniform realization, and irreducible proof-relevant reflective overstructure above bare certification.

## 8.1 Proved tier (complete)

The following are machine-checked with zero proof placeholders: abstract ReflexiveArchitecture with layered coexistence, anti-collapse, three bridge theorems, and the complete non-erasure package; `LinkedArchitecture` with unconditional biconditional; `CrossCorpusInstance` with cross-corpus *Iff* by `Iff.rfl`; concrete source-import instances (`ConcreteFromNEMS`, `ConcreteFromIC`); `nems_spine_both_strata` and `ic_enriched_iff_nems_remainder` (biconditional on the NEMS spine); `universal_nonerasure_law`: the biconditional for all `RolePairDiverseCrownEligible` + `DiagonalCapable` pairs; and `single_hypothesis_nonerasure_law`: the biconditional from the single structural hypothesis `RolePairDiverseCrownEligible` alone (NEMS side always holds from Mathlib's halting problem).

## 8.2 The biconditional and the Universal Non-Erasure Law

*Principle* 8.2 (One-Breath Form of the Non-Erasure Law). In any sufficiently expressive reflective architecture, two conditions are equivalent: there are multiple genuinely distinct proof routes above the canonical certificate, and the architecture's own self-referential truth condition is not computably decidable.

The statements below are the formal machine-checked instantiation of this law. The exact hypotheses and formal objects are given in Theorems 8.3–8.4; the single-hypothesis form appears in Theorem 10.1.

The full logical biconditional between the two strata has been proved, first on the NEMS spine and then in full generality. The key structural predicates are `RolePairDiverseCrownEligible` (IC side: crown-eligible with $\geq 2$ distinct role-separated skeletons, from `infinity-compression-lean`) and `DiagonalCapable` (NEMS side: ASR-equipped, from `nems-lean`).

**Theorem 8.3** (IC Enriched Irreducibility $\iff$ NEMS Semantic Remainder, on NEMS spine)**.** *On the NEMS spine architecture:*

$$\underbrace{\exists m_1 \neq m_2,\ m_1.bare = m_2.bare}_{\textit{IC enriched irreducibility (T-C+.7)}} \iff \underbrace{\neg\,\mathsf{ComputablePred}\ haltingASR.RT}_{\textit{NEMS semantic remainder}}$$

*Lean:* `Bridge.ic_enriched_iff_nems_remainder_unconditional`.

**Theorem 8.4** (Universal Non-Erasure Law)**.** *For any* `RolePairDiverseCrownEligible` *IC architecture $A$ with extraction $H$ and any* `DiagonalCapable` *NEMS framework $F$:*

$$\underbrace{\exists m_1 \neq m_2,\ m_1.bare = m_2.bare}_{\textit{IC enriched irreducibility on } A} \iff \underbrace{\neg\,\mathsf{ComputablePred}\ dc.asr.RT}_{\textit{NEMS semantic remainder on } F}$$

*The NEMS spine biconditional (Theorem 8.3) is a corollary (*`nems_spine_from_universal`*).*
*Lean:* `Bridge.universal_nonerasure_law`, `universal_nonerasure_law_library`, `nems_spine_from_universal`.

**Theorem 8.5** (Both Strata Hold Simultaneously on the NEMS Spine)**.** *On the NEMS spine architecture, both the IC stratum (enriched irreducibility: distinct role-separated skeletons above the same bare certificate) and the NEMS stratum (semantic remainder: record-truth undecidability) hold simultaneously.*
*Lean:* `Bridge.nems_spine_both_strata`, `Bridge.nems_spine_strata_from_sources`.

The two nontrivial implications are: (*i*) *role-pair diversity* $\Rightarrow$ *IC witnesses* via T-C+.7 (from `infinity-compression-lean`); (*ii*) *diagonal capability* $\Rightarrow$ *undecidability* via `diagonal_barrier_rt` (from `nems-lean`). Both sides are structurally forced by their hypotheses independently; the biconditional holds on any pair $(A, F)$ satisfying both. Theorem 8.5 establishes that both strata hold simultaneously on the NEMS spine; Theorem 8.3 establishes their logical equivalence on the spine; Theorem 8.4 lifts this to a universal law over all structurally adequate pairs.

### 8.3 Concrete discharge: `CrossCorpusInstance`

The `CrossCorpusInstance` in `Instances/CrossCorpusInstance.lean` completes the concrete discharge. It builds a `LinkedArchitecture` over carrier `Unit` where $\mathsf{EnrichedIrreducibility}$ is defined as the outer remainder existential *by definition*, so the `enriched_iff_remainder` field is `Iff.rfl`.

**Theorem 8.6** (Cross-Corpus *Iff*)**.** *There exists a concrete* `LinkedArchitecture` *in which the cross-corpus biconditional holds by* `Iff.rfl`*, both coherence axioms are definitional theorems, and the full non-erasure package applies unconditionally.*
*Lean:* `crossCorpusLinkedArchitecture`*,* `crossCorpus_enriched_iff_remainder`*,* `crossCorpus_full_nonerasure`*.*

The unified abstract barrier predicate (`BarrierProp`) simultaneously represents the NEMS non-totalizability content (semantic remainder as $\neg$ FinalSelfTheory) and the IC enriched witness content (enriched irreducibility as $\exists T, \mathsf{SemanticRemainder}(T)$). Setting these equal by definition is the correct abstract bridge: the `ICAntiTranslation` module in `infinity-compression-lean` explicitly states that direct cross-corpus identification requires importing one corpus into the other. The abstract architecture class instead expresses their shared content as a single proposition, making the cross-corpus Iff definitional rather than requiring a new inter-repo theorem.

The concrete instances (EPIC_018) complete the discharge. In `ConcreteFromNEMS.lean`, NEMS record-truth undecidability is the concrete outer semantic remainder: $\mathsf{SemanticRemainder}(-) := \neg\, \mathsf{ComputablePred}\ dc.asr.RT$, supplied directly by `diagonal_barrier_rt`. `ConcreteFromIC.lean` uses T-C+.7 for enriched irreducibility. Zero proof placeholders in both.

## 9 Consequences for the larger program

### 9.1 A formal notion of foundational adequacy

A foundational account is not adequate merely because it is elegant, self-referential, or theorem-rich. It must answer:

1. where determinacy-bearing work lives (outer admissibility),
2. how local structure is globalized via the exact required architecture (middle realization),
3. what reflective structure remains above canonical proof (inner residue),
4. and whether outer and inner non-exhaustion are connected by a provable law (non-erasure biconditional, under the coherence axioms).

The repository `reflexive-architecture-lean` gives this adequacy notion its first explicit machine-checked abstract formal home.

### 9.2 Against flat foundational pictures

Many foundational programs tacitly aim at one of the following ideals: a final external chooser, a universal internal decider, a fixed-point collapse sufficient for all uniform structure, or a canonical theorem that exhausts all reflective content. The combined corpus rejects all four. The non-erasure biconditional sharpens the rejection further: not only can outer and inner structure not be collapsed independently, they are proved equivalent inside ReflexiveArchitecture under the coherence constraints.

### 9.3 A new role for proof architecture

APS and Infinity Compression together suggest that resistant proof targets may fail not only because target statements are hard, but because the proof architecture being used lacks the requisite realization or refinement structure. The Gluing–Route Coherence theorem (Theorem 2.4) makes this explicit: middle gluing data is part of the package that forces the inner fiber structure.

### 9.4 A formal field of architecture theory

The repository `reflexive-architecture-lean` is the first instantiation of a new formal domain: architecture theory for reflexive systems. Its verified lessons inside ReflexiveArchitecture are: semantic burden, realization architecture, and certification residue are irreducible strata of a well-defined abstract class; bare canonicity does not pin reflective finality; middle gluing and inner route discipline interact in a proved theorem; and outer semantic remainder and inner enriched irreducibility are equivalent under the coherence axioms and universal totality.

## 10 Conclusion

**What is proved.** The Strata program is complete. The machine-checked Lean 4 development proves: layered coexistence; stratified non-collapse; three bridge theorems (separation, interaction, independence); the full non-erasure package; the unconditional biconditional in `LinkedArchitecture`; and concrete source-import instances `ConcreteFromNEMS` (barrier = semantic remainder, from `diagonal_barrier_rt`) and `ConcreteFromIC` (enriched irreducibility from T-C+.7). The cross-corpus *Iff* $\mathsf{EnrichedIrreducibility} \iff \exists T, \mathsf{SemanticRemainder}(T)$ holds in `CrossCorpusInstance` by `Iff.rfl`. The joint theorem `nems_spine_both_strata` proves both strata hold simultaneously. The biconditional `ic_enriched_iff_nems_remainder` proves their logical equivalence on the NEMS spine. The Universal Non-Erasure Law `universal_nonerasure_law` generalizes this to all `RolePairDiverseCrownEligible` + `DiagonalCapable` pairs. Zero proof placeholders.

**What this means.** Semantic burden, realization architecture, and reflective certification residue are not merely parallel themes. They are irreducible strata of a common abstract class. The outer and inner strata are linked by a proved unconditional non-erasure biconditional. NEMS record-truth undecidability is the concrete realization of the outer-layer semantic remainder. The law "representation does not erase the represented" is not a philosophical observation; it is a machine-checked theorem that discharges concretely against the actual NEMS and IC source repositories.

**Further strengthening: single-hypothesis form.** `DiagonalCapable` is universally satisfiable: the halting framework (`haltingFramework`, `haltingASR`) is always diagonally capable, independent of any IC architecture. The two-hypothesis universal law therefore collapses to a single hypothesis:

**Theorem 10.1** (Single-Hypothesis Non-Erasure Law)**.** *For any* `RolePairDiverseCrownEligible` *IC architecture A:*

$$\underbrace{\exists m_1 \neq m_2,\ m_1.bare = m_2.bare}_{\textit{IC enriched irreducibility}} \iff \underbrace{\neg\,\mathsf{ComputablePred}\,haltingASR.RT}_{\textit{always true}}$$

*The right-hand side holds unconditionally (Mathlib's halting problem). The law reduces to:* structural IC plurality forces IC enriched irreducibility *(T-C+.7), and NEMS semantic remainder is always available via the canonical halting framework.*

*Lean:* `Bridge.single_hypothesis_nonerasure_law`, `Bridge.single_hypothesis_is_sufficient`.

The biconditional is fundamentally a theorem about IC structural plurality alone. NEMS undecidability — available unconditionally from Mathlib's halting problem — is always the canonical NEMS partner. The program is complete at the abstract, concrete, universal, and single-hypothesis levels.

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